HelioGrid

Hardware & Software • Operation & Maintenance Manual

HelioGrid Energy Management
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Hardware & Software • Operation Maintenance Manual

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For Authorized Administrators Only

Raspbeny Pi 4B-Powered IoT Solutionfor Intelligent Campus Energy Management

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Nominal Capacity	100Ah at 20-hour discharge rate
Full Charge Voltage (Float)	27.2–27.6V (13.6–13.8V per unit at 25°C)
Low Voltage Cutoff	22.5V (bank) — 0% SOC. Restore grid immediately below this level.
Recommended Charge Current	10A (C/10 rate for maximum battery longevity)
Operating Temperature	Discharge: −20°C to +60°C; Charge: 0°C to +45°C
Weight	~28–30 kg per unit
Design Life	3–5 years at 25°C with proper maintenance schedule

12.4 Solar Array Specifications

Parameter	Specification
Panel Rating	590W Monocrystalline
Panel Count	4 panels
Array Configuration	2S2P (2 strings × 2 panels in series)
Total Array Power	2,360W (2.36 kW)
String Voltage (Vmp)	~96V per string
String Voltage (Voc)	~130V per string (maximum open-circuit)
DC Connector	MC4 (IP67, 1000V DC rated)

12.5 Active Sensors Summary

Sensor	Interface / Connection / Normal Range
PZEM-004T #1 (Grid AC)	RS-485 → USB → RPi — Voltage, Current, Power, Frequency — Normal: 210–241V, 59–61Hz
PZEM-004T #2 (Inverter AC)	RS-485 → USB → RPi — Voltage, Current, Power, Frequency — Normal: 210–230V, 59–61Hz
Voltage Divider → Arduino A5	Analog ADC → USB Serial → RPi — Battery DC Voltage — Range: 22.5V (0%) to 25.6V (100%)
INA219 (Battery DC)	I2C → Raspberry Pi GPIO2/GPIO3 — Battery Current (A) and Voltage (V)
DHT22 (Temperature)	GPIO4 → Raspberry Pi — Enclosure Temp: Normal 20°C–40°C, Warning 41–49°C, Critical ≥50°C

Chapter 1 Safety Precautions

Read this manual carefully before operating the HelioGrid device. Follow all safety warnings and instructions to ensure correct, safe installation and operation.

1.1 Safety of Operation

1. Please read this manual carefully before use to ensure correct installation and safe operation.
2. Please pay attention to any warning signs and unusual conditions when using this device.
3. Do not place this device under direct sunlight, rain, or moisture environment.
4. Do not install this device near heat sources, heaters, or furnaces.
5. Install this device in a safe and empty space to ensure ventilation and heat dissipation, and also easy to maintain.
6. Please use dry and insulating rag to clean. Do not use wet cloth or water.
7. In case of fire, please use a dry powder fire extinguisher to put out fire. Liquid fire extinguisher is prohibited.
8. Choose the right space for the device installation. Ensure the battery pack is powerful enough for the system.
9. Make sure the connected appliances and battery capacity match the inverter rated power.

1.2 Safety of Electric

1. Make sure the inverter has been properly grounded and all cables connect in the right socket. Also confirm the battery polarity is in the correct position.
2. To protect the battery, place a circuit breaker with over-current protection between the inverter and battery. This is mandatory.
3. If reconnecting the inverter cable, make sure the inverter is completely shut down and the input breaker and battery switch is off. Failure to follow this procedure may result in electric shock.
4. Never work on live circuits. This system operates at 220–240V AC and 24V DC. Both voltage levels can be lethal.
5. Always use insulated tools rated for at least 1000V AC when working inside the panel.
6. Use a non-contact voltage tester to confirm circuits are de-energized before touching wires.

1.3 Safety of Battery

1. The life span of the battery will be shortened if the environment temperature increases beyond the rated range. Replace the battery regularly to ensure normal inverter operation and sufficient backup power.
2. The replacement and maintenance of the battery must be made by authorized personnel. Use the same type of battery with the same capacity with the same quantity.
3. There are possibilities for electric shock and short-circuit when handling batteries. To avoid that, please follow the instructions below:

A. Please remove your watch, ring, earring, or any metal accessory before handling batteries.

B. Pleaseuseinsulatedtools.

C. Please wear rubber shoes and gloves.

D. Please do not place any metal tools or any metal component on the battery.

E. Please shut down all appliances before removing battery terminals.

F. Non-professionals please do not disassemble the battery or damage the battery. Battery contains dangerous acid which could damage the skin and eyes. If touched accidentally, wash with water and go to hospital for medical examination.

G. Please confirm the battery cable positive and negative terminals before connecting to the battery.

H. Please install a circuit breaker on the battery to prevent fire and electrical shock.

1.4 Operation and Maintenance

1. The operation and storage environment is directly related to the inverter life span and reliability. Therefore, do not place the inverter in the following environments:

A. Temperature/humidity exceeding inverter working environment standard. Inverter should work in 0–55°C, 0–95% humidity environment with no condensation.

B. Anyplacewheretherewillbevibrationandcollision.

C. Any place where there are metallic dust, corrosive substances, salt, or flammable gases.

2. Store the inverter in a dry environment if not in use for a certain amount of time.

3. The environment temperature must be above 0°C for 2 hours before starting up the inverter.

4. Please keep the inverter ventilation holes open so the inverter can ventilate. Insufficient air will cause inner temperature to go up and shorten inner component life span.

5. If not using the battery for a long time, please recharge the battery every 3 months.

⚠ **WARNING:** Power OFF the system and unplug the adapter before handling any component. Failure to do so may result in electric shock, fire, or equipment damage.

Chapter 2 System Components and Functions

The table below lists all hardware and electronic components of the HelioGrid Campus Resilience System with their functions.

Component	Function
Raspberry Pi 4B	Main controller of the entire system. Runs the web dashboard, MariaDB database, sensor polling, anomaly detection engine, relay switching logic, and email alert service.
Arduino Uno	Reads analog sensor data from the voltage divider (battery SOC) and WCS1500 current sensors. Sends data to Raspberry Pi via USB serial.
ONF-SOLAR Hybrid Inverter	3kW hybrid inverter. Handles MPPT solar charging, battery charging, and AC output in both inverter and grid-bypass modes.
Solar PV Array	4 × 590W monocrystalline panels in 2S2P configuration. Provides

Chapter 12 Technical Data Sheet

12.1 System Specifications

Parameter	Specification
System Type	Hybrid Solar + Grid + Battery with IoT Monitoring and Automated Control
Controller	Raspberry Pi 4 Model B — Quad-core 1.5GHz, 4GB LPDDR4
Co-processor	Arduino Uno — ATmega328P, 5V, 16MHz
Database	MariaDB / MySQL (local on Raspberry Pi)
Dashboard URL	http://192.168.254.130
Authentication	Google OAuth 2.0 or Admin Email / Password
Sensor Polling Rate	Every 5 seconds (PZEM); 2-second dashboard refresh
Data Export	CSV download for all sensor logs and anomaly logs
Email Alerts	SMTP via Gmail — configurable from Admin page

12.2 Inverter Specifications — ONF-SOLAR TG-30224M60

Parameter	Specification
Rated Power	3000W (3 kVA)
DC Input Voltage	24V nominal (20–30V range)
Solar MPPT Input	60–145V DC, max 60A MPPT current
AC Output Voltage	220V AC (Philippines)
AC Output Frequency	60Hz (Philippines)
AC Input (Grid Bypass)	90–280V AC, 50/60Hz
Max Battery Charge Current	40A (set to 10A for 100Ah VRLA C/10 rate)
Peak Efficiency	≥93%
Transfer Time (Inverter→Bypass)	< 20ms

12.3 Battery Specifications — JSL II 12V 100Ah VRLA

Parameter	Specification
Type	VRLA Sealed Lead-Acid (maintenance-free)
Configuration	2 units in series = 24V 100Ah bank

10.3 RecommendedNotes

1. Battery: JSL II 12V 100Ah VRLA — match the exact brand and model. Do not mix with other brands.
2. Raspberry Pi: Buy only from authorized Philippine distributors. Avoid counterfeit boards.
3. PZEM-004T: Purchase v3.0 specifically. Earlier versions have different RS-485 behavior.
4. Resistors: Always use 1% tolerance metal film resistors for the voltage divider — 5% carbon film causes inaccurate SOC readings.
5. MicroSD Card: Use only Class 10 / A1 high-endurance cards. Standard cards wear out quickly from continuous database writes.

Chapter 11 Required Tools

Tool	Purpose
Multimeter	Check voltage levels and wire/sensor continuity. Essential for battery, PZEM, and buck converter diagnostics.
Soldering Iron + Solder	Re-solder loose or broken connections on sensor terminals and Arduino pins.
Screwdriver Set (Phillips + Flat)	Remove and install screws on enclosure panels, DIN rail components, and terminal blocks.
Wire Stripper / Cutter	Prepare and replace damaged signal and power wiring.
Heat Shrink Tube + Heat Gun	Insulate newly soldered or reconnected wires.
DC Clamp Meter	Measure PV string current during daylight hours without breaking the circuit.
Non-Contact Voltage Tester	Safely verify live/dead status of AC lines before touching wiring.
Heavy-Duty Grease / Dielectric Grease	Lubricate battery terminals and connector threads to prevent corrosion.
Microfiber Cloth	Clean LCD touchscreen surface and solar panel surfaces.
Laptop / Computer	SSH access to Raspberry Pi for service restart, SD card backup, and log access.
USB-C Cable (Spare, 5V 3A)	Spare power cable for Raspberry Pi in case of cable fault.
Flat-blade Screwdriver (2mm)	Loosen and tighten terminal block screw connections for sensor wires.
Needle-nose Pliers	Handle small components, re-seat connectors, and work in tight enclosure spaces.

	up to 2.36kW of solar input to the inverter MPPT.
Battery Bank (JSL II VRLA)	2 × 12V 100Ah sealed lead-acid batteries wired in series = 24V 100Ah DC storage bank.
PZEM-004T (Grid AC)	Measures grid voltage, current, power, and frequency. Connected via RS-485 to USB adapter to Raspberry Pi.
PZEM-004T (Inverter AC)	Measures inverter AC output voltage, current, power, and frequency. Connected via RS-485 to USB adapter.
Voltage Divider (Battery)	200kΩ / 10kΩ resistor divider across battery terminals. Output feeds Arduino A5 analog pin for battery SOC calculation.
INA219 Module	High-accuracy DC current and voltage sensor. Used for battery bank current monitoring.
WCS1500 Current Sensor (×4)	Hall-effect AC/DC current sensors. Analog output feeds Arduino analog pins A0–A3.
SSR K1 — Solid State Relay	Controls inverter AC output to the load. Active in solar/battery mode.
SSR K2 — Solid State Relay	Controls grid bypass feed to the load. Active in grid bypass mode.
CHNT Contactor	High-current 3-phase contactor for main AC switching. Controlled by Raspberry Pi GPIO.
DHT22 Temperature Sensor	Monitors enclosure ambient temperature and humidity. Connected to Raspberry Pi GPIO4.
DS3231 RTC Module	Real-time clock for accurate event timestamping. Battery-backed (CR2032).
Relay Module	Low-power relay for buzzer activation via Raspberry Pi GPIO signal.
LCD Touchscreen (IPS 5"×7")	Local on-site display for the HelioGrid dashboard. Connected via HDMI and USB touch.
Buzzer	On-site audible alarm for critical anomaly events. Driven by Raspberry Pi GPIO via relay.
Buck Converter	Converts 12V DC to 5V DC. Provides regulated power to the Raspberry Pi USB-C port.
AC MCB Breaker 32A	Protects the inverter AC output lines. (Callout ① in wiring diagram)
DC MCB Breaker 40A	Protects the DC solar input line. (Callout ② in wiring diagram)
MCCB Battery Breaker	Main battery disconnect. Isolates 24V bank from inverter DC bus. (Callout ③)
Grid Breaker	Main grid AC circuit breaker. (Callout ④ in wiring diagram)

Surge Protective Device (SPD)	Protects all components from lightning-induced or switching voltage surges.
Grid Outlet	2-gang outlet on right panel — grid bypass line (K2). (Callout ⑦)
Inverter Outlet	2-gang outlet on right panel — inverter output line (K1). (Callout ⑧)
Battery 24V 100Ah	JSL II 24V 100Ah bank (2 × 12V series). (Callout ⑨ in wiring diagram)
Electrical Moulding	Cable duct protecting wiring runs inside the enclosure. (Callout ⑩)
Core Wires (23AWG / 16AWG)	Sensor signal wiring (23AWG) and power wiring (16AWG+).
MC4 Solar Connectors	Weatherproof DC connectors (1000V rated) for PV panel-to-inverter wiring.

Chapter 3 Installation

3.1 Inspection for Unpacking

1. Open the package. The carton should include the following items:
   1) HelioGrid device set(inverter, control board, battery bank, enclosure)
   2) This Operation & Maintenance Manual
2. Before opening the device package, please check and confirm if the device was damaged during transportation. If any damage or missing parts are found, please contact the installer or distributor immediately.

△ *If any component is missing or damaged, do not attempt to power on the device. Contact the technical team responsible for the installation.*

3.2 Installation Requirements

1. Please make sure a professional electrician installs this inverter. Follow all instructions below:
   1) Do not place anything onto of the inverter.
   2) The installation space must match the inverter size with minimum 30cm clearance on all sides.
   3) Do not install the inverter on a wall made of flammable or heat-resistant material.
   4) Install the inverter as shown in the diagram for easy inspection and maintenance.
   5) Do not install the inverter under direct sunlight.
   6) Installation environment humidity should be 0–95% with no condensation.
   7) Environment temperature should be 0–55°C.
   8) Leave enough space between the enclosure and surrounding walls for air flow.

Battery JSL II 12V 100Ah	JSL II 12V 100Ah VRLA sealed lead acid battery
MC4 Solar Connector	MC4 solar panel connector pair 1000V 30A
USB-C Power Cable	USB-C cable 5V 3A
USB-to-RS485 Adapter	CH340 USB to RS485 adapter module
Relay Module (5V)	5V relay module SPDT SRD-05VDC
Circuit Breaker MCB	MCB 32A DIN rail or 15A MCB DIN rail
Surge Protective Device	SPD surge protector Class II 275V DIN rail
Heat Shrink Tubing	heat shrink tubing assorted set 2:1

10.2 Local Stores — Electrical, Hardware, and Specialty Shops

Store Type	What to Buy Here	*How to Ask*
Electronics Supply Store	PZEM-004T, Arduino Uno, WCS1500, INA219, DHT22, DS3231 RTC, resistors, buck converter, USB-to-RS485 adapters, relay modules	*Ask for microcontroller or IoT sensor components. Bring a photo or spec sheet of the original part.*
Electrical Supply Shop	Circuit breakers (MCB 32A, 40A, 15A), MCCB battery breaker, Surge Protective Device (SPD), DIN rail, terminal blocks, cable duct (electrical moulding), cable lugs	*Specify exact amperage and type (MCB, MCCB). For SPD, say 'Class II 275V DIN rail surge protector.'*
Hardware Store	Wire (16AWG, 23AWG), MC4 solar connectors, cable ties, heat shrink tubing, screws, conduit, electrical tape, dielectric grease	*For wire, specify strand count and insulation type. For solar DC wiring, always ask for UV-resistant outdoor-rated cable.*
Battery Dealer / UPS Shop	JSL II 12V 100Ah VRLA sealed lead-acid batteries	*Bring the old battery to match physical size, terminal type, and Ah rating. Specify 'VRLA sealed, maintenance-free, 12V 100Ah.'*
Computer / IT Store	Raspberry Pi 4B, microSD cards (high-endurance), USB-C cables (5V 3A), HDMI cables	*Buy from trusted authorized resellers only. Avoid clone Raspberry Pi boards.*
Solar Equipment Dealer	MC4 connectors, DC circuit breakers, PV cables (UV-resistant), replacement 590W monocrystalline panels	*Specify 2S2P array and match panel Vmp (~48V) and Isc (~13A) of the original panel.*

Raspberry Pi intermittently loses power or will not boot.	
USB-to-RS485 Adapter — PZEM sensor offline even after cable and port change.	CH340 or CP2102 USB-to-RS485 converter adapter
Relay Module — Buzzer does not sound during critical events.	5V coil SPDT relay module, NO-COM-NC terminals
MC4 Solar Connectors — Visible arcing marks, burnt plastic, or loose engagement.	MC4 pair, 1000V DC rated, 30A current rating
Circuit Breakers (MCB/MCCB) — Breaker trips repeatedly without overload; will not latch.	Match original amperage exactly (32A, 40A, or 15A)
Surge Protective Device (SPD) — Status indicator shows 'faulty' or red.	SPD Class II, 275V, DIN rail mount

Chapter 10 Where to Buy Replacement Parts

10.1 Online Stores (Shopee / Lazada)

Most convenient option. Search for the following keywords:

Component	Search Keyword
PZEM-004T AC Sensor	PZEM-004T v3.0 energy meter module
SSR Relay	solid state relay 25A AC 3-32V DC control
DHT22 Sensor	DHT22 AM2302 temperature humidity sensor
Voltage Divider Resistors	200kΩ 1% metal film resistor, 10kΩ 1% metal film resistor
INA219 Module	INA219 DC current voltage sensor I2C module
WCS1500 Current Sensor	WCS1500 AC DC current sensor module
DS3231 RTC	DS3231 real time clock module I2C
CR2032 Battery	CR2032 3V lithium coin cell battery
Raspberry Pi 4 Model B	Raspberry Pi 4B 4GB (authorized sellers only)
Arduino Uno	Arduino Uno R3 original or compatible
Buck Converter	DC DC step down 12V to 5V 3A adjustable
MicroSD Card (Spare)	microSD card 32GB Class 10 A1 high endurance

△ *Above mentioned requirements are only suitable for ground-mounted or other non-flammable surfaces.*

3.3 Product Overview and Enclosure Layout

The HelioGrid device is installed as a self-contained unit under a solar carport structure. The system consists of two physical assemblies: the solar carport frame with panels above, and the electrical enclosure below.



Figure 3.1 — HelioGrid device installed under the solar carport. 4 × 590W panels mounted above; electrical enclosure mounted at center below.



Figure 3.2 — Enclosure interior view. Left: ONF-SOLAR 3kW Hybrid Inverter + MCCB. Center: Distribution panel (breakers, SSRs, contactor). Right: Raspberry Pi control board. Bottom: JSL II 24V 100Ah battery bank.

Location	Contents
Inverter Bay (Left)	ONF-SOLAR 3kW Hybrid Inverter + Battery MCCB
Distribution Panel (Center)	Circuit breakers (32A×2, 40A, 15A), SPD, SSR K1, SSR K2, CHNT Contactor, Cooling fans (×2)
Control Board (Right-Center)	Raspberry Pi 4B, Arduino Uno, Buck Converter, DS3231 RTC, Relay Module, Buzzer, DHT22
Sensor Board	WCS1500 current sensors (×4), INA219 module, Voltage Divider board (200kΩ/10kΩ)
Outlets (Right Panel)	Grid AC Outlet (K2 — Callout ⑦) and Inverter AC Outlet (K1 — Callout ⑧)
Battery Bay (Bottom)	2 × JSL II 12V 100Ah VRLA in series = 24V 100Ah (Callout ⑨)
Display (Top of Enclosure)	IPS LCD Touchscreen 5"×7" — local HelioGrid dashboard
Side Switch	HelioGrid ON/OFF toggle — controls Raspberry Pi boot and shutdown

3.4 Control Board Layout

The control board houses all electronic components of the HelioGrid monitoring system. The diagram below identifies each component and its location.

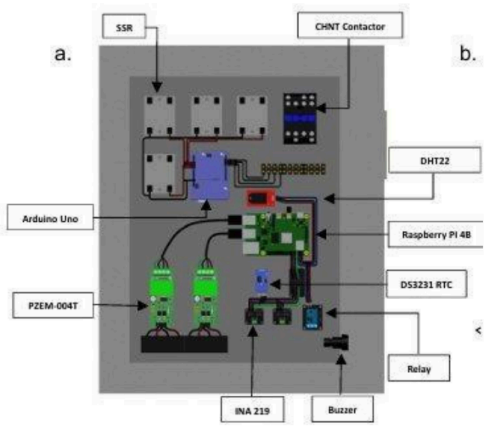


Figure 3.3 — HelioGrid control board layout diagram showing all electronic components and their positions inside the enclosure.

Chapter 9 Replacement Parts — What to Replace and When

The table below lists all replaceable components, the signs that indicate replacement is needed, and the correct replacement specification.

Component — Replace When...	Replacement Specification
PZEM-004TAC Sensor—Offline status, reads 0V, unresponsive after reconnect.	PZEM-004Tv3.0 (80–260V AC, 100A, RS-485 interface)
SSR Relay K1or K2—Relay does not switch; control signal present but no AC switching.	25A AC Solid State Relay, 3–32V DC control input
DHT 22 Sensor—Temperature reads −127°C or 999°C; offline in dashboard.	DHT22/AM2302 temperature and humidity sensor
Voltage Divider Resistors—Battery SOC reads 0% despite battery being charged.	200kΩ and 10kΩ, 1% tolerance metal film resistors
INA266 Module—Battery current reading is zero or random.	INA266 DC current/voltage sensor module (I2C)
WCS1500 Current Sensor—Current reads zero or random despite load present.	WCS1500 ±35 A hall-effect current sensor, 5V supply
Buck Converter—Raspberry Pi will not power on; output <4.8V or absent.	DC-DC step-down module, 12V input→ 5V 3A output
MicroSD Card—Raspberry Pi fails to boot even after re-seating card.	32GB+Class 10/A1 ,high-endurance micro SD card
DS3231 RTC Module—Timestamps wrong; RTC loses time after power cycle.	DS3231 RTC module I2C—replace CR2032 battery first.
CR2032Battery—RTC loses time after power loss.	CR2032 3V lithium coin cell battery
Battery Bank (JSLIIVRLA)—Fails to hold charge after 8–12 hour charging cycle.	JSLII 12V 100Ah VRLA Sealed Lead-Acid (×2 in series for 24V)
USB-C Power Cable—	USB-C cable, 5V 3A rated (minimum 1meter length)

- Wrong charge setting: Adjust inverter battery charge current to 10A in the inverter LCD settings.
- End-of-life: If battery fails to hold charge after proper charging, replace JSL II 12V 100Ah VRLA (×2).

H. Dashboard Not Loading

Diagnostic Steps:

1. Check if the Raspberry Pi power LED is lit. If dark, verify buck converter output and USB-C cable.
2. Try accessing the dashboard from a different device or browser.
3. Check the router DHCP table to confirm Raspberry Pi IP is 192.168.254.130.

Solutions:

- RPi not booted: Power-cycle via HelioGrid side switch. Wait 60seconds be fore accessing.
- Service crash: SSH in and run: sudo system ctl restart heliogrid
- Cache issue: Clear browser cache and retry, or use incognito mode.

Chapter 8 Maintenance Schedule

Frequency	Tasks
Daily	1. Log in and verify 'Normal — System Operational' and 5/5 sensors online. 2. Review Anomaly Alerts for Critical/Warning events. 3. Check battery SOC (should be >43% by end of day). 4. Confirm DHT22 enclosure temperature is 20–40°C.
Weekly	1. Export Anomaly Logs and Energy Logs to CSV for record-keeping. 2. Inspect enclosure exterior for water ingress or physical damage. 3. Clean PV panel surfaces with a soft damp cloth (no water pressure). 4. Check ventilation openings are clear of obstruction.
Monthly	1. Cycle all breakers OFF/ON during a scheduled shutdown. 2. Inspect battery terminals for corrosion — clean and re-tighten. 3. Verify voltage divider wires at battery terminals and Arduino A5. 4. Clear anomaly logs after exporting (Delete All after CSV export).
Quarterly	1. Inspect solar panel mounting and carport for rust or loose bolts. 2. Test on-site buzzer from the System Modules page. 3. Backup Raspberry Pi SD card: sudo dd if=/dev/mmcblk0 of=/backup/heliogrid_backup.img
Annual	1. Full electrical inspection by a licensed electrician. 2. Inspect and test the Surge Protective Device (SPD). 3. Battery capacity test — full discharge/recharge cycle. 4. Re-torque all bus bar and SSR relay terminal connections. 5. Update HelioGrid software if a newer version is available.

Component	Location and Connection
SSR (Solid State Relay)	Top-left of board. AC IN/OUT terminals. Control signal from Raspberry Pi GPIO.
Arduino Uno	Left-center of board. USB serial connection to Raspberry Pi. Reads A0–A5 analog inputs.
PZEM-004T (×2)	Bottom-left of board. RS-485 to USB adapter → Raspberry Pi USB port. CT clamp on AC lines.
Buzzer	Bottom-right of board. Driven by Relay module via Raspberry Pi GPIO.
DHT22	Right-top of board. Data wire → Raspberry Pi GPIO4. 10kΩ pull-up to 3.3V.
Raspberry Pi 4B	Right-center of board. Main controller. USB-C power from Buck Converter (5V 3A).
DS3231 RTC	Right-center of board. I2C connection to Raspberry Pi. CR2032 backup battery.
Relay Module	Right-bottom of board. 5V coil. Drives buzzer via Raspberry Pi GPIO output.

3.5 Electrical Wiring Diagram—Numbered Parts

The diagram below shows the complete power circuit wiring of the HelioGrid system. Each numbered callout corresponds to a major component in the system.

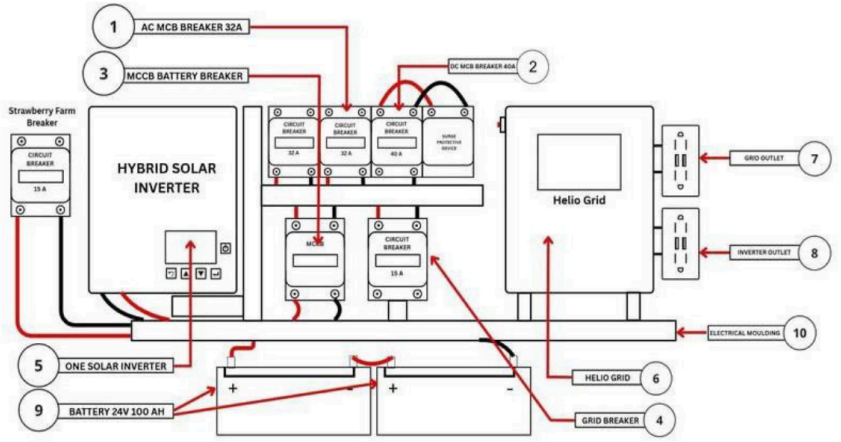


Figure 3.4 — HelioGrid Electrical Wiring Diagram. All numbered callouts are identified in the parts list below.

Parts Identification:

No.	Name	Specification	Function
①	AC MCB Breaker 32A	*MCB 32A, DIN Rail, Single-pole*	Protects the inverter AC output line. This is the red-handle MCB connected directly to inverter AC output.
②	DC MCB Breaker 40A	*MCB 40A, DIN Rail, DC-rated*	Protects the DC solar PV input line between the panels and the inverter MPPT terminal.
③	MCCB Battery Breaker	*MCCB (Molded Case Circuit Breaker)*	Main battery disconnect. Isolates the 24V 100Ah battery bank from the inverter DC bus. Must be OFF before any battery maintenance.
④	Grid Breaker	*MCB 15A, Strawberry Farm Breaker*	Main grid AC circuit breaker on the grid feed line from the farm supply. Serves as the main grid ON/OFF switch in start-up and shutdown procedures.
⑤	ONE Solar Inverter	*ONF-SOLAR TG-30224M60, 3kW, 24V*	Hybrid solar inverter. Manages MPPT charging from solar, battery charging from grid, and AC output to loads in both solar and bypass modes.
⑥	Helio Grid	*Raspberry Pi 4B Control Box*	The HelioGrid IoT control box. Contains Raspberry Pi 4B, Arduino Uno, sensors, relays, and all monitoring and control electronics.
⑦	Grid Outlet	*2-gang outlet, 15A, 250V rated*	Grid bypass power outlet mounted on the right panel. Connected to the grid AC line via SSR K2 relay.
⑧	Inverter Outlet	*2-gang outlet, 15A, 250V rated*	Inverter power outlet mounted on the right panel. Connected to inverter AC output via SSR K1 relay.
⑨	Battery 24V 100Ah	*JSL II 12V 100Ah VRLA × 2 (series)*	Two sealed lead-acid batteries connected in series to form the 24V 100Ah DC storage bank. Supplies power during solar-only operation and grid outages.
⑩	Electrical Moulding	*PVC cable duct, 25mm or 40mm*	Cable duct that organizes and protects internal wiring runs. Keeps high-voltage and low- voltage wiring separated for safety.

- Loose A5 wire: Re-solder or reconnect. Verify with continuity check.
- Calibration offset: Adjust Arduino sketch voltage calibration constant to match multimeter reading.

E. Raspberry Pi or Arduino Does Not Boot

Symptoms:

- Nodisplayoutput.Dashboardnotaccessiblewithin2minutes.
- Sensorsshownodataevenwhensystemappearspoweredon.

Diagnostic Steps:

1. Verify Raspberry Pi power: 5V 3A via USB-C (from buck converter). Arduino: 5V 2A.
2. Try a different USB-C cable for the Raspberry Pi.
3. Press the microSD card firmly to ensure proper seating. If still failing, remove and test the card.
4. Check the USB cable from Arduino to Raspberry Pi — try a different USB port.
5. Inspect the Micro-HDMI to HDMI cable between the Raspberry Pi and the LCD touchscreen.

Solutions:

- Corrupted SD card: Re-flash the HelioGrid OS image to a new high-endurance micro SD card.
- Service crash: SSH in and run: sudo system ctl restart heliogrid
- Network issue: Check router DHCP table for actual Raspberry Pi IP address.

F. Solar Input — No Generation

Diagnostic Steps:

1. Check the solar circuit breaker (DC MCB 40A — ②) — must be ON.
2. Inspect solar panel surfaces — clean if dusty, shaded, or covered with debris.
3. Measure PV string open-circuit voltage with a DC multimeter — expected ~130V OC for 2S array.
4. Inspect MC4 connectors at panels and at inverter MPPT input for proper engagement.

Solutions:

- Dirty panels: Clean with a soft damp cloth. Never use pressurized water.
- Disconnected MC4: Re-engage. Replace damaged MC4 connectors if arcing mark are visible.
- Low PV voltage (<60V): Check for open - circuit in one panel—inspect individual panel wiring.

G. Battery — Drains Quickly or Does Not Charge

Diagnostic Steps:

1. Check the 220V AC input at the grid connection and the Grid Farm Breaker (④).
2. Measure battery voltage at terminals — below 22.5V indicates deep discharge.
3. Verify inverter charge current setting in LCD menu — should be 10A for 100Ah VRLA.
4. Check the battery MCCB (③) — must be ON to allow inverter to charge the battery.

Solutions:

- Deep discharge: Connect grid power immediately. Allow 8–12 hours of charging at 10A before use.

B. Buck Converter or Raspberry Pi Does Not Power On

Symptoms:

- Raspberry Pi has no power LED. Dashboard not accessible after switchis turned ON.
- No display output on LCD touchscreen.

Diagnostic Steps:

1. Measure buck converter output voltage with a multimeter — should read 5.0–5.2V DC.
2. Inspect the USB-C cable from buck converter to Raspberry Pi for damage or loose fit.
3. Check the 12V input to the buck converter — should read 11.5–12.5V from battery tap.

Solutions:

- Loose connection: Re-seat USB-C cable and verify buck converter input wires.
- Incorrect voltage: Adjust buck converter potentiometer to 5.1V DC output.
- Faulty module: Replace with DC-DC step-down module (12V→5V, 3A rated minimum).

C. PZEM-004T Sensor — Offline or Reading Zero

Symptoms:

- Sensor shows 'Offline' (reddot) in dashboard Active Sensors.
- Grid or Inverter voltage shows 0Vdespite power being present.

Diagnostic Steps:

1. Check USB-to-RS485 adapter at the Raspberry Pi USB port — try a different USB port.
2. Verify device path: run ls /dev/tty* and confirm /dev/ttyGridPZEM and /dev/ttyInverterPZEM appear.
3. Inspect PZEM-004T current transformer (CT) clamp — must be clamped around only one AC conductor.
4. Check terminal connections at PZEM-004T RS-485 port for loose wires.

Solutions:

- Disconnected USB: Reconnect and run: sudo system ctl restart heliogrid
- Wrong device path: Reassign USB port mapping in HelioGrid configuration file.
- Faulty PZEM: Replace with PZEM-004Tv3.0 module (80–260VAC, 100A, RS-485).

D. Battery SOC Reading Incorrect (Voltage Divider)

Symptoms:

- Battery SOC reads 0% when battery is clearly charged.
- Dashboard battery voltage does not match multi-meter reading at terminals.

Diagnostic Steps:

1. Measure battery voltage at terminals with a multimeter — note the actual reading.
2. Measure voltage at divider output (wire going to Arduino A5) — should be ≈ battery voltage × 10k / 210k.
3. Verify resistors: 200kΩ high-side and 10kΩ low-side with multimeter in resistance mode.
4. Inspect the output wire to Arduino A5 for loose connection or short to GND.

Solutions:

- Wrong resistor: Replace with 200kΩ and 10kΩ, 1% metal film resistors.

3.6 Electronic Schematic

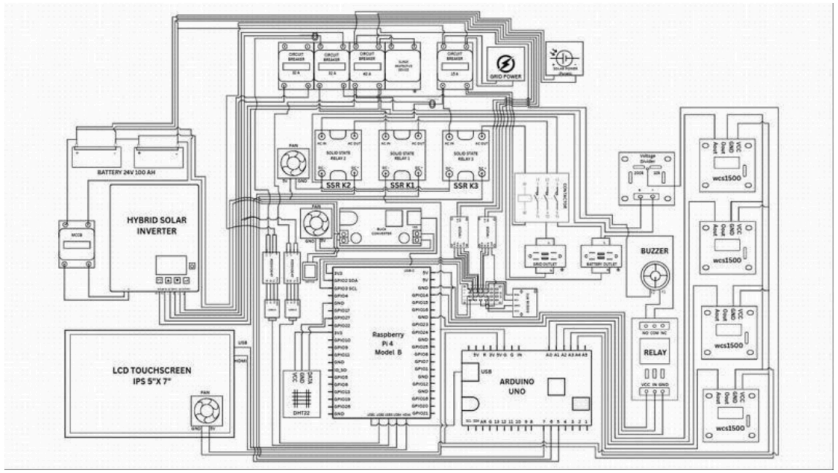


Figure 3.5 — Full System Electronic Schematic. Shows Raspberry Pi 4B GPIO connections, PZEM-004T RS-485 sensors, Arduino Uno analog inputs (WCS1500, voltage divider A5), INA219 I2C bus, SSR K1/K2, CHNT contactor, buzzer relay, buck converter, DHT22, and DS3231 RTC.

3.7 Key Wiring Connections

Connection	Specification
Battery (Series)	Red (+) Battery 1 → Black (−) Battery 2 = 24V output. MCCB (③) on positive output lead.
Solar Input	2S2P: 2 strings × 2 × 590W panels → Inverter MPPT. Protected by DC MCB 40A (②).
Inverter AC Output	Inverter → SSR K1 → Inverter Outlet (⑧). Controlled by Raspberry Pi GPIO.
Grid AC Input	Grid → Strawberry Farm Breaker (④) → SPD → 40A main bar → SSR K2 → Grid Outlet (⑦). Controlled by Raspberry Pi GPIO.
Voltage Divider	200kΩ (high-side) + 10kΩ (low-side) across battery terminals → Arduino A5 analog input.
PZEM-004T #1 (Grid)	RS-485 → USB adapter → Raspberry Pi. CT clamp on grid AC line. Path: /dev/ttyGridPZEM
PZEM-004T #2 (Inverter)	RS-485 → USB adapter → Raspberry Pi. CT clamp on inverter

	output line. Path: /dev/ttyInverterPZEM
INA219	SDA → Raspberry Pi GPIO2; SCL → GPIO3. Monitors battery bank current and voltage.
WCS1500 (×4)	In-line on AC load and bypass lines. Aout → Arduino analog pins A0–A3. Powered from 5V rail.
DHT22	Data → Raspberry Pi GPIO4. 10kΩ pull-up resistor to 3.3V.
Arduino → Raspberry Pi	USB-A (Arduino) → USB port on Raspberry Pi for serial data.
Buck Converter	Input: 12V (battery tap) → Output: 5V → Raspberry Pi USB-C power port.
Buzzer	Driven by Raspberry Pi GPIO via relay module. 5V power supply from buck converter.

Chapter 4 Operating

4.1 DashboardOverview

Item	Detail
Dashboard URL	http://localhost:3000
Authentication	Google OAuth (Continue with Google) or admin email and password
Access Level	Authorized Administrators Only
Auto-Refresh	Every 2 seconds (live data)
Admin Email	smartenergycontrol0@gmail.com

4.2 SystemModes

The HelioGrid system operates in the following modes based on energy availability and threshold rules:

Mode	Description
Solar Priority (Auto)	Solar energy powers the load first. Surplus charges battery. Grid used only when solar is insufficient.
Grid Bypass (Auto)	Grid powers the load when solar is insufficient or a critical anomaly is detected. SSR K2 active.
Battery Priority	Battery and solar power loads independently. Grid is not used. For grid-outage scenarios.
Hybrid Auto (HelioGrid)	HelioGrid software auto-switches K1/K2 relays based on real-time threshold monitoring. Default mode.

Step	Action	Details
1	**HelioGrid Switch OFF**	Flip HelioGrid (⑥) side switch to OFF. Wait 30 seconds for Raspberry Pi to complete graceful shutdown (status LED goes dark).
2	**Inverter MCB OFF**	Switch OFF the red inverter MCB 32A (①). Inverter Outlet (⑧) is now de-energized.
3	**Inverter OFF**	Press the ONF-SOLAR Inverter power button to turn OFF. Wait for display to go dark.
4	**Solar Breaker OFF**	Switch OFF the blue solar circuit breaker (DC MCB 40A — ②). PV array disconnected from MPPT.
5	**Battery MCCB OFF**	Switch OFF the battery MCCB (③). 24V battery bank now isolated from inverter.
6	**Grid Farm Breaker OFF**	Switch OFF the 15A Strawberry Farm Breaker (Grid Breaker ④).
7	**Grid Breaker OFF**	Switch OFF the blue grid breaker. System is now fully de-energized.

After shutdown: confirm all breaker handles are in the OFF (down) position. Lock the enclosure door when leaving site unattended.

Chapter 7 Troubleshooting

> ⚠ **WARNING:** Power OFF the system and unplug the adapter before handling any component.

A. SSR Relay — Load Does Not Switch or System Stuck on Bypass

Symptoms:

- System remain song rideven when solar/battery is healthy.
- Relay does not respond when anomaly threshold is triggered.

Diagnostic Steps:

1. Power OFF the system.
2. Locate SSR K1, K2 inside the distribution panel.
3. Check the control signal wire from Raspberry Pi GPIO to the SSR input terminal.
4. Use a multimeter to verify 3–5V DC control signal at SSR input during a triggered event.
5. Inspect SSR input and output terminals for loose, burnt, or corroded connections.

Solutions:

- Loose wire: Tighten or solder the connection.
- No control signal: Verify Raspberry Pi GPIO pin assignment in HelioGrid software.
- Faulty SSR: Replace with equivalent 25A AC solid state relay (same AC IN/OUT voltage rating).

3. Confirm ALL circuit breaker handles are in the OFF (down) position.
4. Ensure Raspberry Pi is connected to power via Buck Converter → USB-C cable.

⚠ **WARNING:** Perform steps EXACTLY in the order listed. Do not skip or reorder any step.

Step	Action	Details
1	**Solar Breaker ON**	Turn ON the blue solar circuit breaker (DC MCB 40A — ②). Connects PV array to inverter MPPT input.
2	**Battery MCCB ON**	Switch ON the battery MCCB (③) to enable 24V DC source. Battery indicator activates on inverter display.
3	**Inverter ON**	Long-press the ONF-SOLAR Inverter power button (~3 seconds) until display activates. Wait for startup chime.
4	**Stabilization Wait**	Wait 10–15 seconds for inverter output to stabilize at 220–230V, 60Hz. Observe inverter LCD display.
5	**Inverter MCB ON**	Turn ON the red inverter MCB 32A (①). Inverter AC Outlet (⑧) is now live.
6	**Grid Farm Breaker ON**	Switch ON the 15A Strawberry Farm Breaker (Grid Breaker ④) on the left side of the panel.
7	**Grid Breaker ON**	Switch ON the blue grid breaker to connect grid AC to the bypass circuit.
8	**HelioGrid Switch ON**	Flip the HelioGrid (⑥) side switch to ON. Raspberry Pi boots in 30–60 seconds. Dashboard: http://192.168.254.130

After start-up, verify the following on the dashboard:
1. Energy Flow Hub shows 'Auto' mode active.
2. SOLAR node shows voltage (>0V during daylight) and status ACTIVE.
3. Battery shows SOC percentage and voltage between 24–27.6V.
4. Inverter shows ~220–223V AC output. Status: INVERTER or BYPASSED.
5. System Status shows: Normal — System Operational.
6. Active Sensors shows 5/5 Online.

6.2 ShutdownSequence

⚠ **WARNING:** Never cut main power abruptly.This corrupts the Raspberry Pi SD card and may damage inverter settings.

4.3 Monitoring Thresholds and Automated Responses

Condition	Rule and Automated Response
Grid Normal	210–241V, 59–61 Hz — continue current mode, log event.
Grid Warning	200–209V or 241–245V — dashboard alert only. No buzzer, no email.
Grid Critical	<200V or >245V, or |Δf|>1Hz — switch to bypass; Buzzer 5s; email alert.
Grid Dropout	V=0V or V<23V — immediate Critical, no delay.
Voltage Spike	|ΔV|≥8V → Warning; |ΔV|≥23V outside 207–253V → Critical.
Voltage Drift	1V < |ΔV| < 7V → Warning; ≥2 readings outside 207–253V → Critical.
Battery 0% SOC	≤22.5V — restore grid power immediately.
Sensor Freeze	>15s no data → default to grid (K2 ON), Buzzer 5s, suspend switching, email alert.
Temperature Warning	41°C – 49°C (DHT22) — dashboard alert only.
Temperature Critical	≥50°C (DHT22) — email alert; investigate enclosure immediately.

4.4 Dashboard Screen Guide

Login Screen

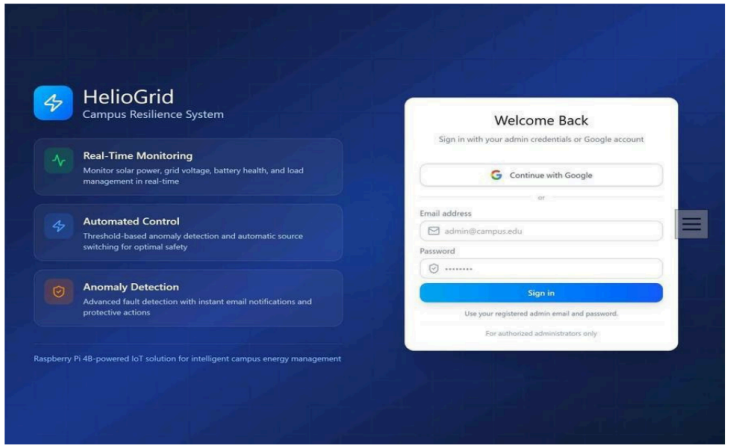


Figure 4.1 — Login screen. Sign in using Google OAuth or registered admin email and password. For authorized administrators only.

Energy Flow Hub — Home

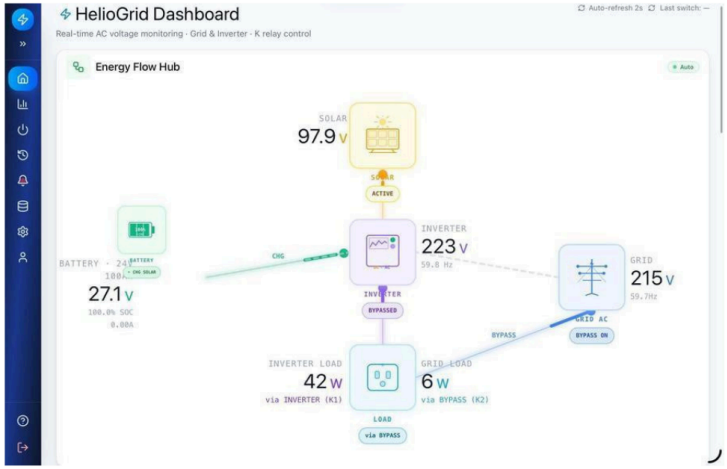


Figure 4.2 — Energy Flow Hub. Live visualization: Solar (97.9V active), Battery (27.1V, 100% SOC), Inverter (223V bypassed), Grid (215V bypass ON). Load split: K1 = 42W, K2 = 6W. 'Auto' badge = automatic relay control active.

Real-Time Monitoring

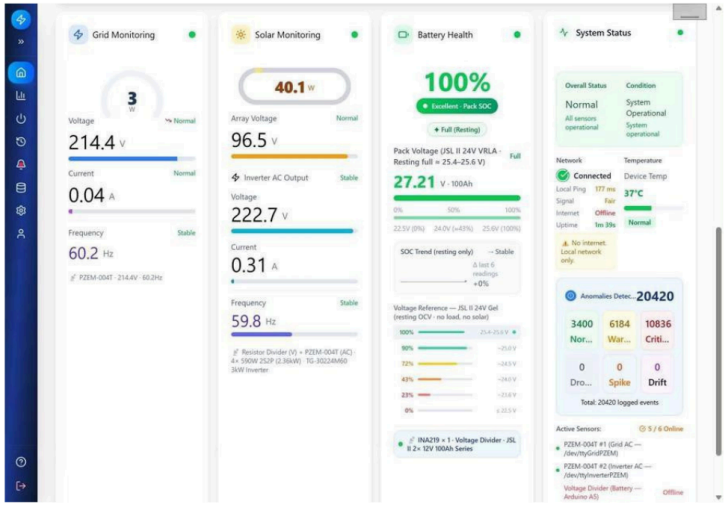


Figure 4.3 — Real-time monitoring panels: Grid (214.4V, 60.2Hz, Normal), Solar (40.1W, 96.5V), Battery (100% SOC, 27.21V, Full-Resting), and System Status (sensor count, anomaly totals, DHT22 temperature).

• Work with dry hands in a dry environment near electrical components.	• Operate the system with the enclosure door open in rain or high humidity.

5.2 Battery Handling

✓ DO	✗ DO NOT
• Use both hands when handling batteries — each unit weighs approximately 28–30 kg.	• Short battery terminals — even momentarily. Causes arcing, fire, or explosion.
• Verify correct polarity: Red = (+), Black = (−) before connecting cables.	• Tilt or tip VRLA batteries — internal acid may leak.
• Ensure battery MCCB (③) is OFF before connecting or disconnecting terminals.	• Charge above 10A (C/10 rate for 100Ah battery).
• Keep terminals clean and apply dielectric grease to prevent corrosion.	• Mix batteries of different brands, ages, or capacities in the same bank.
• Check battery resting voltage ≥ 12.5V per unit before installation.	• Expose batteries to temperatures above 40°C or open flame.

5.3 Solar PV and Inverter

✓ DO	✗ DO NOT
• Work on the solar array at night or when all panels are fully covered.	• Disconnect MC4 connectors under load — causes dangerous arcing.
• Use DC-rated MC4 connectors (≥600V DC) for all PV cable connections.	• Step on, stand on, or drop tools onto solar panels.
• Allow 10–15 seconds for inverter to stabilize after power-on before connecting the load MCB.	• Connect load MCB before inverter has fully stabilized — causes voltage spikes.
• Follow Start-Up and Shutdown sequences in exact order.	• Service internal inverter components — contact manufacturer.
• Record any inverter fault codes shown on the display before resetting.	• Operate the inverter with the battery MCCB switched off.

Chapter 6 Operating Procedures

The HelioGrid device must be started and shut down in the correct sequence. Skipping or reordering steps may cause relay conflicts, inverter faults, or SD card corruption.

6.1 Start-UpSequence

Before starting, confirm all of the following:

1. Visually inspect the enclosure — no visible damage, moisture, or loose wiring.
2. Confirm battery voltage on inverter display is ≥ 22.5V.

Admin Account

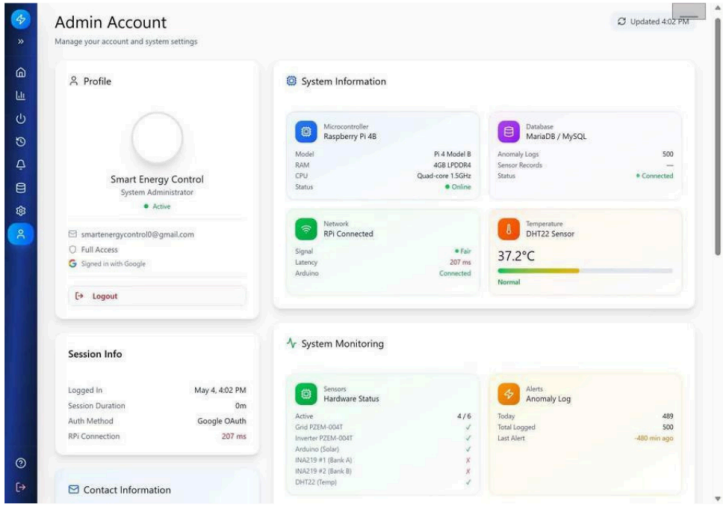


Figure 4.10 — Admin Account. Shows Raspberry Pi 4B hardware info, MariaDB status, network latency, DHT22 temperature (37.2°C Normal), session details (Google OAuth), and individual hardware sensor status.

Chapter 5 Electrical Do's and Don'ts

These rules apply to all operators, technicians, and administrators at all times when working with or around the HelioGrid device.

5.1 General Electrical

✓ DO	✗ DO NOT
• De-energize the full system using the Shutdown Sequence before opening the enclosure.	• Open the enclosure while the system is energized.
• Use insulated tools rated for at least 1000V AC when working inside the panel.	• Work on the system alone — always have a second person present.
• Wear PPE: insulated gloves, safety glasses, and non-conductive footwear.	• Use bare-metal or uninsulated probes near live terminals.
• Use a non-contact voltage tester to confirm de-energization before touching wires.	• Bypass any circuit breaker, fuse, or relay (SSR K1, K2, CHNT contactor).
• Ensure the enclosure is properly grounded before energizing.	• Remove or short-circuit the Surge Protective Device (SPD).

Real-Time Analytics

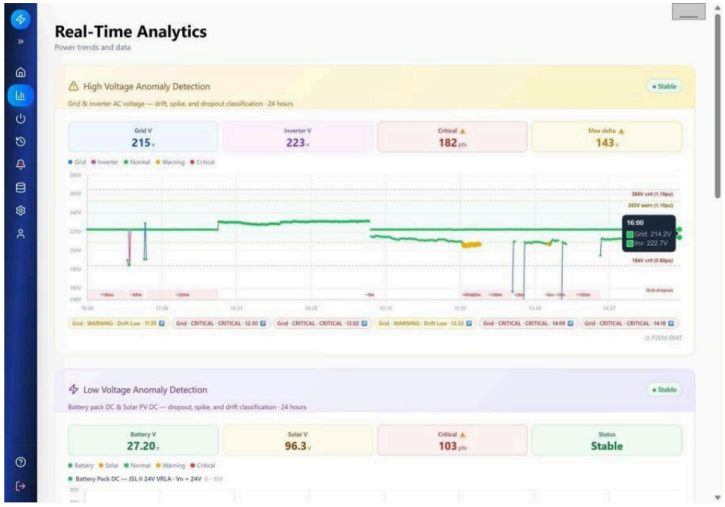


Figure 4.4 — High Voltage Anomaly Detection: 24-hour chart of grid and inverter AC voltage with critical threshold bands (265V, 184V). Drift, spike, and dropout events shown with timestamps.



Figure 4.5 — Voltage Trends (24h), Current Trends (24h), and Cumulative Energy (kWh) charts for Grid (blue) and Inverter (pink) channels.

Anomaly Detection Logs

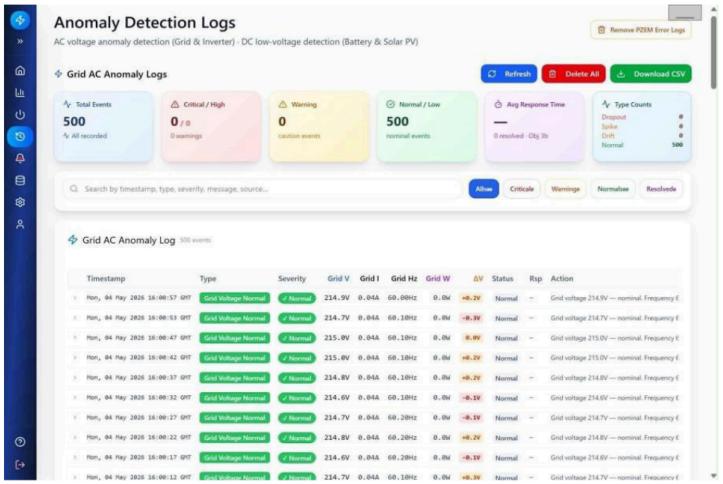


Figure 4.6 — Grid AC Anomaly Log. Columns: Timestamp, Event Type, Severity, Grid Voltage (V), Current (A), Frequency (Hz), Power (W), Delta-V (ΔV), Status, Response time, Action. Downloadable as CSV.

Anomaly Alerts and Notifications

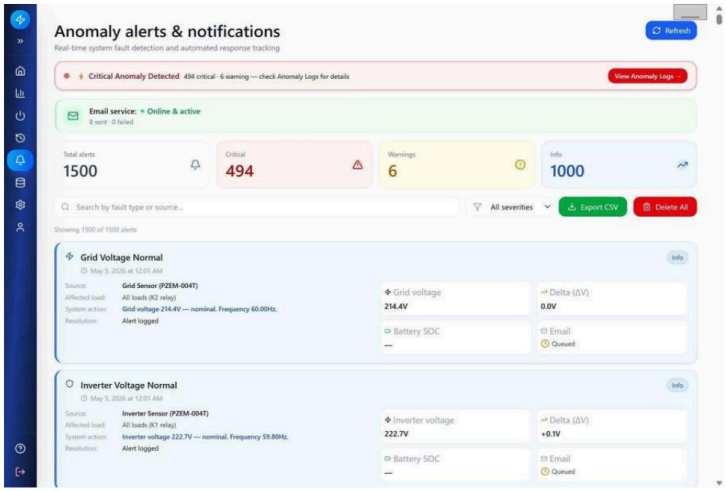


Figure 4.7 — Anomaly Alerts. 1,500 total events: 494 Critical, 6 Warning, 1,000 Info. Email service: Online & active (8 sent). Each alert shows source sensor, affected load, system action, voltage, SOC, ΔV, and email delivery status.

Energy and Network Logs

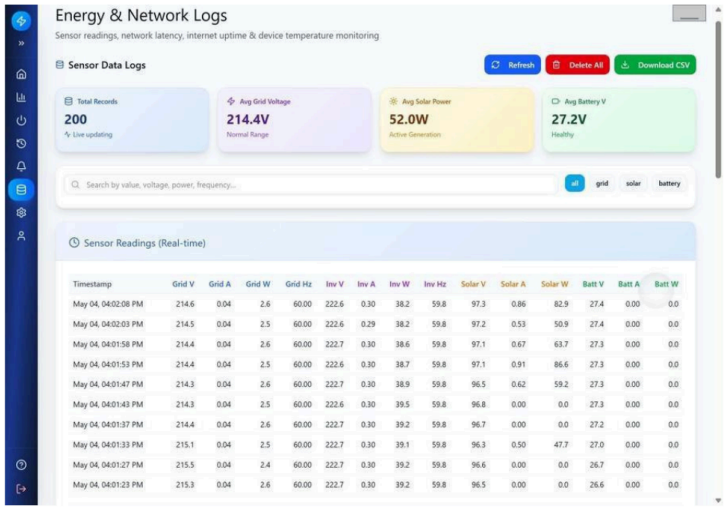


Figure 4.8 — Sensor Data Logs table. All readings with timestamp: Grid V/A/W/Hz, Inverter V/A/W/Hz, Solar V/A/W, Battery V/A/W. Averages shown in header. Downloadable as CSV.

System Modules and Rules

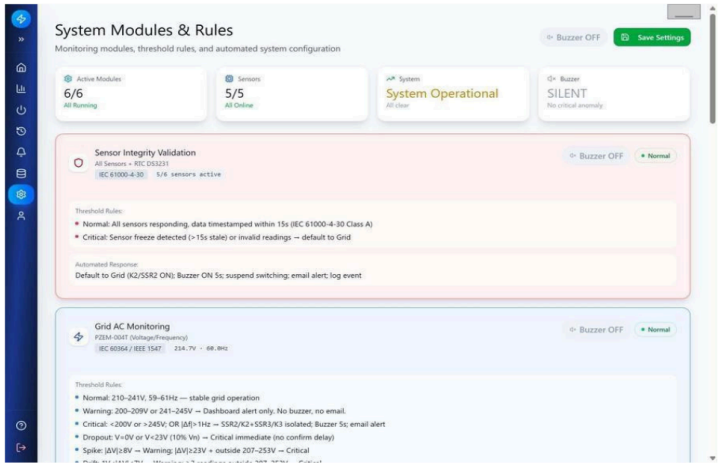


Figure 4.9 — System Modules: 6/6 modules active, 5/5 sensors online. Configurable threshold rules per IEC 60364 / IEEE 1547 / IEC 61000-4-30. Buzzer toggle per module. Save Settings button.